Tutorial 3: Logistic Regression in R

Dataset: Polish Companies Bankruptcy - 4year.arff

Software: R (Posit.cloud)

By the end of the tutorial, you should be able to

- Import (and convert to csv) an ARFF file and identify what one observation represents.
- Validate a binary bankruptcy outcome and inspect class imbalance and missingness.
- Estimate logistic-regression models using interpretable financial ratios.
- Interpret coefficients, odds ratios, predicted probabilities and average marginal effects.
- Apply the model checks introduced in Lecture 3: McFadden’s pseudo-R^2, confusion matrix, threshold choice, ROC curve and AUC.
- Communicate firm-level associations to a bank credit-risk or financial-stability audience without making unsupported causal claims.

Files used

File	Purpose
4year.arff	Original UCI data: fourth-year financial ratios with bankruptcy status two years later.
polish_bankruptcy_codebook.xlsx	Definitions of X1-X64, outcome coding, missing values and dataset provenance.
tutorial03_starter.Rmd	R Markdown file containing prompts and incomplete code chunks.

Core variables used in the tutorial

Renamed variable	Original	Meaning
bankrupt	class	1 = company became bankrupt within two years; 0 = company remained operating.
profitability	X1	Net profit / total assets.
leverage	X2	Total liabilities / total assets.
working_capital	X3	Working capital / total assets.
current_ratio	X4	Current assets / short-term liabilities.
asset_turnover	X9	Sales / total assets.
log_assets	X29	Logarithm of total assets; a proxy for firm size.

Part A. Import and validate the ARFF file

```
install.packages(c("foreign", "dplyr", "ggplot2", "broom",
                   "marginaleffects", "pscl", "pROC"))

library(foreign)
library(dplyr)
library(ggplot2)

raw <- read.arff("data/4year.arff")
dim(raw)
names(raw)
head(raw)

# The file contains 64 ratios followed by the class label.
names(raw)[1:64] <- paste0("X", 1:64)
names(raw)[ncol(raw)] <- "bankrupt"

table(raw$bankrupt, useNA = "ifany")
prop.table(table(raw$bankrupt))
colSums(is.na(raw))
```

- *What does one row represent?*
- *Are the only outcome values 0 and 1?*
- *What proportion of observations belongs to the bankruptcy class?*
- *Why is class balance especially important before using a confusion matrix?*

Part B. Select and prepare the modelling variables

```
df <- raw %>%
  transmute(
    bankrupt = as.integer(as.character(bankrupt)),
    profitability = X1,
    leverage = X2,
    working_capital = X3,
    current_ratio = X4,
    asset_turnover = X9,
    log_assets = X29
  )

summary(df)
colSums(is.na(df))

model_df <- na.omit(df)
nrow(df) - nrow(model_df)
table(model_df$bankrupt)
prop.table(table(model_df$bankrupt))
```

- *How many observations were removed by complete-case analysis?*
- *Could missing-value deletion change the represented sample?*
- *Why should the outcome remain numeric 0/1 for this exercise?*

Part C. Explore the outcome and predictors

```
model_df %>%
  group_by(bankrupt) %>%
  summarise(across(profitability:log_assets,
                   list(mean = mean, median = median)))

ggplot(model_df, aes(x = factor(bankrupt), y = profitability)) +
  geom_boxplot() +
  labs(x = "Bankruptcy status", y = "Net profit / total assets")

ggplot(model_df, aes(x = factor(bankrupt), y = leverage)) +
  geom_boxplot() +
  labs(x = "Bankruptcy status", y = "Total liabilities / total assets")
```

- *Which ratios show the clearest raw differences?*
- *Why do raw group differences not establish that a ratio causes bankruptcy?*

Part D. Estimate logistic-regression models

```
model1 <- glm(
  bankrupt ~ profitability,
  data = model_df,
  family = binomial(link = "logit")
)

model2 <- glm(
  bankrupt ~ profitability + leverage + working_capital +
    current_ratio + asset_turnover + log_assets,
  data = model_df,
  family = binomial(link = "logit")
)

summary(model1)
summary(model2)
```

- *Which coefficients are positive or negative?*
- *For H0: beta_j = 0, which predictors have p-values below 0.05?*
- *Why should a logit coefficient not be interpreted directly as a probability-point change?*
- *Does statistical significance establish causality?*

Part E. Convert results into interpretable quantities

```
library(broom)
library(marginaleffects)

# Odds ratios
odds_ratios <- tidy(model2) %>%
  mutate(odds_ratio = exp(estimate))
odds_ratios
```

```
# Average marginal effects in probability units
ame <- avg_slopes(model2, type = "response")
ame

# Predicted probabilities for all observations
model_df$predicted_probability <- predict(model2, type = "response")
summary(model_df$predicted_probability)
```

- *Interpret one odds ratio while holding the other included variables constant.*
- *Interpret one average marginal effect in probability or percentage-point terms.*
- *Why is an odds ratio not the same as a probability ratio?*
- *Which measure is clearest for a non-technical banking audience?*

Part F. McFadden’s pseudo-R^2

```
library(pscl)
pR2(model2)["McFadden"]
```

- *Do not interpret pseudo-R^2 as the percentage of variation explained.*
- *What does it compare the fitted model with?*
- *How does it help describe model fit without proving that the model is correct?*

Part G. Confusion matrix and threshold choice

```
classification_metrics <- function(actual, probability, threshold) {
  predicted <- ifelse(probability >= threshold, 1, 0)
  cm <- table(Predicted = predicted, Actual = actual)

  TN <- cm["0", "0"]
  FN <- cm["0", "1"]
  FP <- cm["1", "0"]
  TP <- cm["1", "1"]

  c(
    threshold = threshold,
    accuracy = (TP + TN) / sum(cm),
    precision = TP / (TP + FP),
    recall = TP / (TP + FN)
  )
}

p_hat <- predict(model2, type = "response")

classification_metrics(model_df$bankrupt, p_hat, 0.50)
classification_metrics(model_df$bankrupt, p_hat, 0.10)

table(Predicted = ifelse(p_hat >= 0.10, 1, 0),
      Actual = model_df$bankrupt)
```

- *How does lowering the threshold change precision and recall?*
- *Why can accuracy be misleading when bankruptcy is uncommon?*

- *In a bank-screening context, which is more costly: a false negative or a false positive?*
- *Why should the threshold reflect the intended application?*

Part H. ROC curve and AUC

```
library(pROC)

roc_object <- roc(model_df$bankrupt, p_hat, levels = c(0, 1))
auc(roc_object)

plot(roc_object,
    main = "ROC curve: Polish-company bankruptcy model",
    col = "black",
    lwd = 2)
abline(a = 0, b = 1, lty = 2, col = "grey50")
```

- *What does an AUC of 0.50 represent?*
- *Does the model discriminate better than chance?*
- *Why does a high AUC not prove causality or correct probability estimates?*

Part I. Communicate the evidence

Prepare a 180-word briefing for a bank credit-risk or financial-stability team. Include:

- the dataset, forecasting horizon and unit of observation;
- the bankruptcy rate and one important descriptive pattern;
- one adjusted association expressed using an odds ratio or marginal effect;
- the McFadden pseudo-R^2, AUC and chosen threshold;
- one limitation and one appropriate next analysis;
- a clear statement that the results are associations and screening evidence, not causal effects.

Data Source

Tomczak, S. (2016). Polish Companies Bankruptcy [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5F600 (CC BY 4.0).